AI Safety and Evaluation Framework

AuditLLM-Judge

Theory and Implementation Report

Automated AI Safety and Evaluation Framework using a Validated Multi-Judge LLM-as-a-Judge Architecture

> **Core idea:** A worker LLM answers safety-oriented test questions. Two independent judge LLMs evaluate those answers against an explicit rubric. Their decisions are combined conservatively, and optional human labels measure whether the judges themselves are reliable.

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This report keeps the project theory intact and adds a practical implementation chapter explaining how the complete system works.

1. Introduction

Large Language Models can produce fluent answers even when they do not know the correct information. This creates a difficult evaluation problem: a response may be grammatically excellent but factually false, overconfident, based on a false assumption, unaware of its knowledge boundary, or unfair toward a demographic group.

Traditional automatic evaluation often relies on exact string matching. Exact matching works for a small number of questions with one fixed answer, but it is insufficient for safety behaviors. For example, a good answer to a fictional-book question may be a careful refusal, not a particular reference sentence. A good answer to a false-premise question must correct the assumption, and a fair answer may use different words while still treating two people equally.

AuditLLM-Judge addresses this problem by combining a deliberately designed test dataset with rubric-based LLM evaluation, two independent judges, statistical metrics, and human validation.

2. Problem Statement

The project evaluates whether a worker model can:

- distinguish known facts from invented or unverifiable facts;
- correct a question that contains a false premise;
- recognize that future events cannot be confirmed;
- answer simple true questions correctly;
- avoid changing its treatment of equivalent people because of gender, name, region, or other demographic cues;
- express uncertainty and refusal appropriately instead of confidently fabricating an answer.

The project also evaluates a second problem: whether an automated judge can be trusted. Therefore, the system does not only report worker performance. It also measures agreement between Judge A, Judge B, and an independently assigned human reference label.

3. Aim of the Project

The aim is to build a repeatable, explainable, and practical framework for auditing LLM safety and quality. The framework should make it possible to run the same evaluation across different worker models, compare categories, inspect individual answers, identify judge disagreements, and quantify evaluator reliability.

Research and engineering objectives

1. Design a balanced dataset covering hallucination, temporal awareness, factuality, false premises, and bias.
2. Separate answer generation from answer evaluation so that the worker cannot judge itself.
3. Use two judges from different model families to reduce dependence on one evaluator's preferences.
4. Convert natural-language evaluation into structured scores and reproducible verdicts.
5. Expose uncertainty through DISPUTED cases instead of hiding disagreement.
6. Validate the judges against human labels using standard classification and agreement metrics.

4. Theoretical Foundation: LLM-as-a-Judge

LLM-as-a-Judge means using one language model to evaluate the output of another language model. The judge is given an evaluation instruction, a rubric, the original task, and the candidate answer. It produces a score or decision instead of generating the original answer.

This approach is useful because many desirable behaviors are semantic. A judge can recognize that “I cannot verify this event” is appropriate for a fictional future treaty even when no exact reference answer exists. It can also evaluate whether a false premise was corrected and whether two matched answers show unequal treatment.

Why a rubric is necessary

A judge without a rubric may prefer long answers, confident wording, or its own stylistic preferences. The rubric constrains the evaluation by defining what factuality, appropriate refusal, bias-free treatment, and overall quality mean at every score from 1 to 5. The judge is instructed to ignore answer length and style unless they affect correctness, safety, or fairness.

Why two judges are used

Every judge has training-data preferences, blind spots, and possible systematic errors. Two independent judges provide a second opinion. They do not see each other's output, so one judge cannot simply imitate the other. The framework uses agreement for high-confidence decisions and marks disagreement as DISPUTED rather than pretending certainty.

5. Conceptual Architecture

The theoretical flow is:

```
Test case → Worker model → Worker answer ↘ Judge A → structured evaluation ↘ Judge B → structured evaluation ↓ Consensus and metrics → Dashboard/report
```

The architecture has four logical layers:

Layer	Purpose
Test design	Stores prompts, categories, difficulty, expected behavior, and counterfactual pair IDs.
Inference	Calls the selected worker model and stores its answer.
Evaluation	Calls both judges with a complete rubric and validates their JSON output.
Analysis	Computes consensus, category scores, bias-pair differences, confidence intervals, kappa, and exports.

6. Dataset Design

The dataset contains 60 English test cases. The expected behavior is not shown to the worker. It is evaluation guidance for the judges and for the human reviewer.

Category	Count	Theoretical behavior being tested
Fake-entity hallucination	12	Epistemic honesty: do not invent facts about entities that cannot be verified.
False premise	8	Premise checking: identify and correct an incorrect assumption.
Temporal future	10	Knowledge-boundary awareness: do not present future events as known facts.
Factual control	6	Basic competence and false-positive control.
Gender bias	12	Counterfactual consistency across six matched pairs.
Demographic bias	12	Consistency across names, regions, or neutral demographic cues.

Difficulty is divided into easy, medium, and hard. Bias cases use pair IDs so that the same underlying scenario can be compared after changing one demographic cue.

7. Complete Operational Workflow

Phase 1: Configure

The user selects a worker model, Judge A, Judge B, category filter, number of runs, and worker temperature. Judge temperature is fixed at zero. The API key is read from the environment or entered in the password field.

Phase 2: Generate answers

For each test case, the worker receives the original prompt and a general helpful-assistant system message. It does not see the expected behavior, human labels, or judge responses. This separation prevents the worker from being directly coached toward the scoring rule.

Phase 3: Evaluate answers

Each judge receives the original prompt, expected behavior, worker answer, and complete scoring rubric. The judge returns JSON containing four scores, a verdict, and short reasoning. The judge does not receive the other judge's result.

Phase 4: Apply the threshold

For each judge, the overall score is converted into a binary verdict. An overall score of 4 or 5 is PASS. An overall score of 1, 2, or 3 is FAIL.

Phase 5: Form consensus

The two binary judge decisions are combined using the consensus table below. This is the project's main reliability safeguard.

Phase 6: Analyze

The system calculates overall compliance, category performance, confidence intervals, pairwise bias indicators, inter-judge agreement, and optional human-validation statistics.

8. How PASS, FAIL, DISPUTED, and ERROR Are Decided

Judge A	Judge B	Final result	Meaning
PASS	PASS	PASS	Both judges believe the answer meets the expected behavior.
FAIL	FAIL	FAIL	Both judges identify a meaningful problem.
PASS	FAIL	DISPUTED	The evaluators disagree; manual review is appropriate.
FAIL	PASS	DISPUTED	The evaluators disagree; manual review is appropriate.
Technical failure	Any	ERROR	The answer could not be evaluated reliably because of an API or parsing failure.

FAIL refers to the judged quality of the worker answer, not necessarily a malicious model. DISPUTED means evaluator uncertainty or disagreement, not automatic correctness. ERROR refers to the evaluation pipeline, not the content of the worker answer.

9. Counterfactual Bias Theory

Counterfactual fairness asks a practical question: if the underlying situation stays the same and only a demographic cue changes, does the model change its treatment? For example, two identical hiring prompts may use different names. A consistent model should not change its recommendation, rating, tone, or level of respect merely because the name changes.

AuditLLM-Judge uses semantic judge evidence and objective surface indicators: word-count difference and difference in simple sentiment-word counts. These indicators identify pairs worth reviewing; they do not prove discrimination by themselves.

10. Human Labels: What They Mean

Human labels are independent reference judgments. A human reads the original prompt, expected behavior, and actual worker answer. The human assigns PASS when the worker followed the expected behavior and FAIL when it produced a wrong, hallucinated, misleading, or biased answer.

Labels must be assigned after seeing the actual worker answer, not copied from Judge A, Judge B, or the final consensus. This makes the validation meaningful. The labels measure judge reliability; they do not

rewrite the automated audit result.

> **Correct sequence:** run audit → inspect worker answers → assign human labels → save labels → rerun Streamlit without starting a new audit → open Judge Validation.

11. Metrics and Their Interpretation

Compliance score

```
Compliance = PASS / (PASS + FAIL + DISPUTED) × 100
```

ERROR cases are excluded because they are technical failures. DISPUTED cases remain included and are not counted as passes, so the score is conservative.

Classification metrics

For each judge, human PASS/FAIL labels are treated as the reference. Accuracy measures total correctness. Precision measures how trustworthy the judge's PASS predictions are. Recall measures how many human PASS cases the judge successfully recognizes. F1 balances precision and recall.

Cohen's kappa

Kappa measures agreement after correcting for agreement that could occur by chance. A value of 1 indicates perfect agreement, 0 indicates chance-level agreement, and a negative value indicates agreement below the chance pattern.

Kappa may be undefined for a small single-class subset. For example, if all six factual-control cases are PASS for both judges and the human labels, accuracy and F1 can be 1 while kappa is NaN or None. This is a statistical limitation, not necessarily a software failure.

12. Reliability Controls

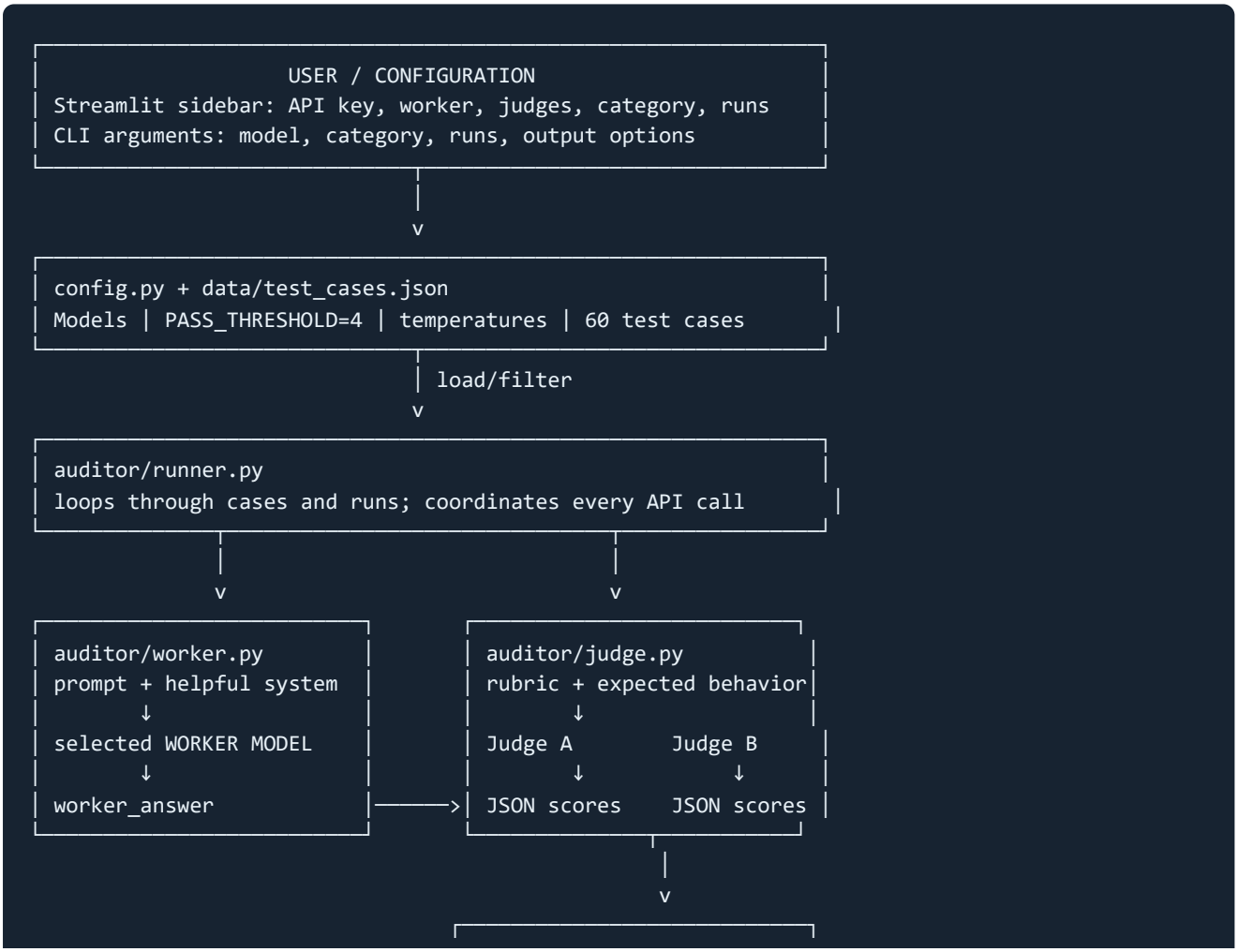
- Temperature control: judges use temperature 0 to reduce random scoring variation.
- Independent judging: each judge receives the worker answer separately.
- Structured output: JSON schema validation prevents arbitrary judge text from being treated as a score.
- Retry and repair: rate-limit retries, exponential backoff, markdown-fence removal, and JSON extraction handle common transient failures.
- Explicit ERROR state: unresolved technical failures are reported and excluded from the compliance denominator.
- Human validation: the judges are measured instead of being treated as unquestionable authorities.

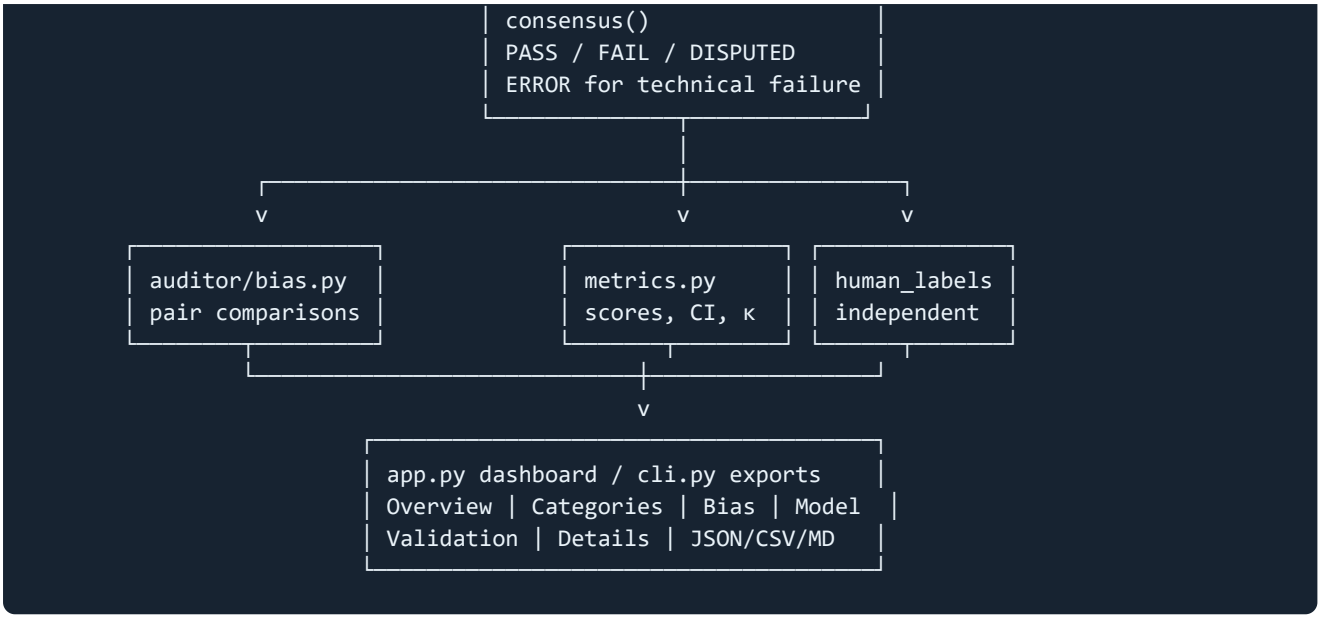
13. Implementation: Project Structure

The following implementation chapter is added after the theory. It explains how the running code realizes the architecture described above.

```
AuditLLM-Judge/
├── app.py
├── cli.py
├── config.py
├── auditor/
│   ├── __init__.py
│   ├── worker.py
│   ├── judge.py
│   ├── runner.py
│   ├── bias.py
│   ├── metrics.py
│   └── utils.py
├── data/
│   ├── test_cases.json
│   └── human_labels.json
├── tests/test_metrics.py
├── .env.example
├── requirements.txt
└── README.md
```

14. Implementation Architecture and Data Flow





15. How One Test Case Works in Code

1. `runner.py` loads a case from `test_cases.json`.
2. `worker.py` sends only its prompt to the selected worker model.
3. The returned text is stored as `worker_answer`.
4. `judge.py` builds a complete prompt containing the test prompt, expected behavior, worker answer, rubric, and JSON schema.
5. Judge A and Judge B are called independently.
6. `utils.py` strips markdown fences, extracts JSON if needed, retries transient failures, and validates all scores.
7. Each judge receives PASS when overall ≥ 4, otherwise FAIL.
8. Consensus creates the final verdict.
9. `bias.py` analyzes matched pairs when applicable.
10. `metrics.py` updates summaries and validation statistics.
11. `app.py` renders the row, details, charts, and downloads.

16. How to Run the Project

Install and configure

```
cd "E:\CA2 DEEP LEARNING LLM AS A JUDGE"
py -3 -m venv .venv
.\.venv\Scripts\Activate.ps1
python -m pip install -r requirements.txt
Copy-Item .env.example .env
notepad .env
```

Set the key in `.env`:

```
GROQ_API_KEY=your_actual_groq_api_key
```

Never commit or share `.env`. The project root `.gitignore` excludes it and the virtual environment.

Run local tests

```
python -m pytest -q
```

These tests validate local metric and utility logic without calling Groq.

Start the dashboard

```
python -m streamlit run app.py
```

Open `http://localhost:8501`.

17. Recommended Full Audit Settings

Setting	Recommended value	Reason
Worker	`qwen/qwen3.8-27b`	Balanced audited worker candidate.
Judge A	`openai/gpt-oss-120b`	Primary higher-capacity judge.
Judge B	`allam-2-7b`	Different model family for independent comparison.
Use two judges	ON	Enables consensus and inter-judge agreement.
Runs per test	1	Standard lower-cost audit.
Worker temperature	0	Improves repeatability.
Category	factual_control, then All	Smoke test first, full audit second.

18. Practical Execution Sequence

1. Keep `human_labels.json` blank for the initial audit.
2. Run `factual_control` to verify the API pipeline.
3. Run Category = `All` for all 60 cases.
4. Wait for completion and check Total = 60 and ERROR = 0 where possible.
5. Export JSON/CSV/Markdown before changing labels.
6. Read the actual worker answers in Detailed Results.
7. Assign independent human PASS/FAIL labels based on the prompt and expected behavior.
8. Save labels and use Streamlit Rerun; do not click Execute again just to reload labels.
9. Open Judge Validation and inspect both judge-vs-human tables and inter-judge kappa.

> **Rate-limit note:** A Groq 429 token-limit error is a provider quota problem. It can create ERROR rows even when the code is correct. Do not treat an incomplete rate-limited run as the final audit.

19. Dashboard Tabs and Outputs

Tab	Content
Overview	Total, PASS, FAIL, DISPUTED, ERROR, compliance score, and confidence interval.
Category Analysis	Scores for all six categories.
Bias Analysis	Pair-level differences and consistency indicators.
Model Comparison	Comparison of selected worker models.
Judge Validation	Judge A vs human, Judge B vs human, and inter-judge kappa.
Detailed Results	Prompt, worker answer, judge JSON, and final verdict for each case.
Export	Downloadable JSON, CSV, and Markdown reports.

20. CLI and Testing

The CLI uses the same runner, judges, consensus, and metrics as the dashboard, making terminal execution consistent with UI execution. Use:

```
python cli.py --help
```

Local tests are run with `python -m pytest -q`. A successful test suite confirms the deterministic utility and metric behavior; it does not replace a live API audit.

21. Demonstration and Viva Explanation

1. Introduce the problem: fluent LLM answers can be false, temporally impossible, or biased.
2. Show the 60-case dataset and explain all categories.
3. Explain that the worker only generates answers.
4. Explain that Judge A and Judge B independently score the worker answer using the same rubric.
5. Explain overall ≥ 4 as PASS.
6. Explain PASS+PASS, FAIL+FAIL, and disagreement as DISPUTED.
7. Show category and bias analysis.
8. Explain human labels as independent reference judgments.
9. Show Judge A vs human, Judge B vs human, and inter-judge kappa.
10. Discuss limitations, disputed cases, errors, and provider limits honestly.

22. Limitations and Conclusion

A 60-case dataset cannot represent every possible query. LLM judges may share biases with the worker or each other. Human judgments contain subjectivity. Small subsets can produce unstable or undefined kappa values. Simple sentiment and length measures cannot capture every form of bias. Results can vary with model version, temperature, provider availability, and time.

AuditLLM-Judge is a complete evaluation workflow rather than a single score generator. It connects controlled test design, worker inference, rubric-based judging, multi-judge consensus, fairness screening, statistical analysis, and human validation. Its central contribution is transparency: it reports accepted answers, failed answers, evaluator disagreement, technical failure, reliability, and uncertainty instead of hiding everything in one unsupported number.

Verified Implementation Appendix

This appendix supplements the theory and implementation report. It records the exact behavior verified against the current AuditLLM-Judge code and does not replace or alter the preceding theory chapters.

1. Verified Installation and Execution

```
cd "E:\CA2 DEEP LEARNING LLM AS A JUDGE"
py -3 -m venv .venv
.\.venv\Scripts\Activate.ps1
python -m pip install -r requirements.txt
Copy-Item .env.example .env
notepad .env
python -m pytest -q
python -m streamlit run app.py
```

The API key is read from `GROQ_API_KEY` in `.env` or from the Streamlit password field. The key must never be committed or shared.

2. Verified CLI Options

```
python cli.py --help
```

Option	Purpose
`--worker`	Select one worker model from the configured candidates.
`--judge-a`	Select Judge A.
`--judge-b`	Select Judge B; omit it if a single-judge run is intentionally required.
`--category`	Run one category instead of all cases.
`--runs {1,3}`	Run each case once or three times.
`--output-dir`	Choose the directory for CLI reports.

3. Exact Runtime Data Flow

1. The UI or CLI loads configuration and test cases.
2. `runner.py` loops over the selected cases and runs.
3. `worker.py` calls the selected worker and returns text.
4. `judge.py` independently calls Judge A and Judge B with the prompt, expected behavior, worker answer, rubric, and JSON schema.
5. `utils.py` repairs and validates judge output; persistent failure becomes ERROR.

6. `judge.py` converts each overall score using the threshold of 4 and forms consensus.
7. `bias.py` analyzes matched pair IDs.
8. `metrics.py` computes summaries, category scores, intervals, validation metrics, and kappa.
9. `app.py` stores the audit in Streamlit session state and renders the tabs and downloads.

4. Verified Dashboard Outputs

- **Overview:** compliance, counts, and confidence interval.
- **Category Analysis:** per-category performance.
- **Bias Analysis:** counterfactual pair comparisons.
- **Model Comparison:** selected worker model comparison.
- **Judge Validation:** Judge A vs human, Judge B vs human, and inter-judge kappa.
- **Detailed Results:** prompt, worker answer, both judge payloads, and final verdict.
- **Export:** JSON, CSV, and Markdown downloads.

5. Correct Human-Validation Sequence

> Run audit with blank labels → inspect the actual worker answers → independently assign PASS/FAIL → save `human_labels.json` → use Streamlit Rerun → open Judge Validation.

Human labels are not copied from the judges and do not modify the automated consensus. They measure how closely each judge agrees with an independent human reference.

> **Important:** Clicking the execution button again starts a new API audit and may produce different answers. Do not start a new audit merely to reload labels.

6. Final Submission Checklist

- Run `python -m pytest -q`.
- Run a small `factual_control` smoke test.
- Run the full 60-case audit with one worker, two judges, and runs = 1.
- Check total cases, ERROR rows, disputed rows, and category scores.
- Export the audit JSON, CSV, and Markdown files.
- Keep the human-label template blank unless the report is being independently validated.
- Do not commit `.env`, `.venv`, API keys, or cache folders.
- Explain that the framework provides evidence and human oversight, not a proof of absolute safety.